

List of mathematical logic topics

3 languages

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This is a **list of mathematical logic topics**.

For traditional **syllogistic logic**, see the list of topics in logic. See also the list of computability and complexity topics for more theory of algorithms.

Working foundations [edit]

- Peano axioms
 - Giuseppe Peano
- Mathematical induction
 - Structural induction
 - Recursive definition
- Naive set theory
 - Element (mathematics)
 - Ur-element
 - Singleton (mathematics)
 - Simple theorems in the algebra of sets
- Algebra of sets
 - Power set
 - Empty set
 - Non-empty set
 - Empty function
- Universe (mathematics)
- Axiomatization
- Axiomatic system
 - Axiom schema
- Axiomatic method
- Formal system
- Mathematical proof
 - Direct proof
 - Reductio ad absurdum
 - Proof by exhaustion
 - Constructive proof
 - Nonconstructive proof
- Tautology
- Consistency proof
- Arithmetization of analysis
- Foundations of mathematics
- Formal language
- Principia Mathematica*
- Hilbert's program
- Impredicative
- Definable real number
- Algebraic logic
 - Boolean algebra (logic)
- Dialectica space
- categorical logic

Model theory [edit]

- Finite model theory
 - Descriptive complexity theory
 - Model checking
 - Trakhtenbrot's theorem
- Computable model theory
 - Tarski's exponential function problem
 - Undecidable problem
- Institutional model theory
 - Institution (computer science)
- Non-standard analysis
 - Non-standard calculus
 - Hyperinteger
 - Hyperreal number
 - Transfer principle
 - Overspill
 - Elementary Calculus: An Infinitesimal Approach
 - Criticism of non-standard analysis
 - Standard part function
- Set theory
 - Forcing (mathematics)
 - Boolean-valued model
- Kripke semantics
 - General frame
- Predicate logic
 - First-order logic
 - Infinitary logic
 - Many-sorted logic
 - Higher-order logic
 - Lindström quantifier
 - Second-order logic
- Soundness theorem
- Gödel's completeness theorem
 - Original proof of Gödel's completeness theorem
- Compactness theorem
- Löwenheim–Skolem theorem
 - Skolem's paradox
- Gödel's incompleteness theorems
- Structure (mathematical logic)
- Interpretation (logic)
- Substructure (mathematics)
- Elementary substructure
 - Skolem hull
- Non-standard model
- Atomic model (mathematical logic)
- Prime model
- Saturated model
- Existentially closed model
- Ultraproduct
- Age (model theory)
 - Amalgamation property
 - Hrushovski construction
- Potential isomorphism
- Theory (mathematical logic)
 - Complete theory
 - Vaught's test
 - Morley's categoricity theorem
 - Stability spectrum
 - Morley rank
 - Stable theory
 - Forking extension
 - Strongly minimal theory
 - Stable group
 - Tame group
 - o-minimal theory
 - Weakly o-minimal structure
 - C-minimal theory
 - Spectrum of a theory
 - Vaught conjecture
 - Model complete theory
 - List of first-order theories
 - Conservative extension
 - Elementary class
 - Pseudoelementary class
 - Strength (mathematical logic)
- Differentially closed field
- Exponential field
- Ax–Grothendieck theorem
- Ax–Kochen theorem
- Peano axioms
- Non-standard model of arithmetic
- First-order arithmetic
- Second-order arithmetic
- Presburger arithmetic
- Wilkie's theorem
- Functional predicate
- T-schema
- Back-and-forth method
- Barwise compactness theorem
- Skolemization
- Lindenbaum–Tarski algebra
- Löb's theorem
- Arithmetical set
- Definable set
- Ehrenfeucht–Fraïssé game
- Herbrand interpretation / Herbrand structure
- Imaginary element
- Indiscernibles
- Interpretation (model theory) / Interpretable structure
- Pregeometry (model theory)
- Quantifier elimination
- Reduct
- Signature (logic)
- Skolem normal form
- Type (model theory)
- Zariski geometry

Set theory [edit]

- Algebra of sets
- Axiom of choice
 - Axiom of countable choice
 - Axiom of dependent choice
 - Zorn's lemma
- Boolean algebra (structure)
- Boolean-valued model
- Burali-Forti paradox
- Cantor's back-and-forth method
- Cantor's diagonal argument
- Cantor's first uncountability proof
- Cantor's theorem
- Cantor–Bernstein–Schröder theorem
- Cardinality
 - Aleph number
 - Aleph-null
 - Aleph-one
 - Beth number
 - Cardinal number
 - Hartogs number
- Cartesian product
- Class (set theory)
- Complement (set theory)
- Complete Boolean algebra
- Continuum (set theory)
 - Suslin's problem
- Continuum hypothesis
- Countable set
- Descriptive set theory
 - Analytic set
 - Analytical hierarchy
 - Borel equivalence relation
 - Infinity-Borel set
 - Lightface analytic game
 - Perfect set property
 - Polish space
 - Prewellordering
 - Projective set
 - Property of Baire
 - Uniformization (set theory)
 - Universally measurable set
- Determinacy
 - AD+
 - Axiom of determinacy
 - Axiom of projective determinacy
 - Axiom of real determinacy
- Empty set
- Forcing (mathematics)
- Fuzzy set
- Internal set theory
- Intersection (set theory)
- L
- L(R)
- Large cardinal property
- Musical set theory
- Ordinal number
 - Infinite descending chain
 - Limit ordinal
 - Successor ordinal
 - Transfinite induction
 - ∈-induction
 - Well-founded set
 - Well-order
- Power set
- Russell's paradox
- Set theory
 - Alternative set theory
 - Axiomatic set theory
 - Kripke–Platek set theory with urelements
 - Morse–Kelley set theory
 - Naive set theory
 - New Foundations
 - Positive set theory
 - Zermelo–Fraenkel set theory
 - Zermelo set theory
- Set (mathematics)
- Simple theorems in the algebra of sets
- Subset
- ℔ (set theory)
- Tree (descriptive set theory)
- Tree (set theory)
- Union (set theory)
- Von Neumann universe
- Zero sharp

Descriptive set theory [edit]

- Analytical hierarchy

Large cardinals [edit]

- Almost Ramsey cardinal
- Erdős cardinal
- Extendible cardinal
- Huge cardinal
- Hyper-Woodin cardinal
- Inaccessible cardinal
- Ineffable cardinal
- Mahlo cardinal
- Measurable cardinal
- N-huge cardinal
- Ramsey cardinal
- Rank-into-rank
- Remarkable cardinal
- Shelah cardinal
- Strong cardinal
- Strongly inaccessible cardinal
- Subtle cardinal
- Supercompact cardinal
- Superstrong cardinal
- Totally indescribable cardinal
- Weakly hyper-Woodin cardinal
- Weakly inaccessible cardinal
- Woodin cardinal
- Unfoldable cardinal

Recursion theory [edit]

- Entscheidungsproblem
- Decision problem
- Decidability (logic)
- Church–Turing thesis
- Computable function
 - Algorithm
 - Recursion
 - Primitive recursive function
 - Mu operator
 - Ackermann function
 - Turing machine
 - Halting problem
 - Computability theory, computation
 - Herbrand Universe
 - Markov algorithm
 - Lambda calculus
 - Church–Rosser theorem
 - Calculus of constructions
 - Combinatory logic
 - Post correspondence problem
- Kleene's recursion theorem
- Recursively enumerable set
 - Recursively enumerable language
- Decidable language
- Undecidable language
- Rice's theorem
- Post's theorem
- Turing degree
- Effective results in number theory
- Diophantine set
- Matijasevič's theorem
- Word problem for groups
- Arithmetical hierarchy
 - Subrecursion theory**
 - Presburger arithmetic
 - Computational complexity theory
 - Polynomial time
 - Exponential time
 - Complexity class
 - Complexity classes P and NP
 - Cook's theorem
 - List of complexity classes
 - Polynomial hierarchy
 - Exponential hierarchy
 - NP-complete
 - Time hierarchy theorem
 - Space hierarchy theorem
- Natural proof
- Hypercomputation
 - Oracle machine
- Rózsa Péter
- Alonzo Church
- Emil Post
- Alan Turing
- Jacques Herbrand
- Haskell Curry
- Stephen Cole Kleene
- Definable real number

Proof theory [edit]

- Metamathematics
- Cut-elimination
- Tarski's undefinability theorem
- Diagonal lemma
- Provability logic
- Interpretability logic
- Sequent
- Sequent calculus
- Analytic proof
- Structural proof theory
- Self-verifying theories
- Substructural logics
 - Structural rule
 - Weakening
 - Contraction
 - Linear logic
 - Intuitionistic linear logic
 - Proof net
 - Affine logic
 - Strict logic**
 - Relevant logic
- Proof-theoretic semantics

- Ludics
- System F
- Gerhard Gentzen
- Gentzen's consistency proof
- Reverse mathematics
- Nonfirstorderizability
- Interpretability
- Weak interpretability
- Counterinterpretability
- Tolerant sequence
- Cotolerant sequence**
- Deduction theorem
- Cirquent calculus

Mathematical constructivism [edit]

- Nonconstructive proof
- Existence theorem
- Intuitionistic logic
- Intuitionistic type theory
- Type theory
- Lambda calculus
 - Church–Rosser theorem
- Simply typed lambda calculus
- Typed lambda calculus
- Curry–Howard isomorphism
- Calculus of constructions
- Constructivist analysis
- Lambda cube
- System F
- Introduction to topos theory
- LF (logical framework)
- Computability logic
- Computable measure theory
- Finitism
- Ultrafinitism
- Luitzen Egbertus Jan Brouwer

Modal logic [edit]

- Kripke semantics
- Sahlqvist formula
- Interior algebra

Theorem provers [edit]

- First-order resolution
- Automated theorem proving
- ACL2 theorem prover
- E equational theorem prover
- Gandalf theorem prover**
- HOL theorem prover
- Isabelle theorem prover
- LCF theorem prover
- Otter theorem prover
- Paradox theorem prover**
- Vampire theorem prover
- Interactive proof system
- Mizar system
- QED project
- Rocq, **formerly Coq**

Discovery systems [edit]

- Automated Mathematician
- Eurisko

Historical [edit]

- Begriffsschrift*
- Systems of Logic Based on Ordinals* – Alan Turing's Ph.D. thesis

See also [edit]

- Kurt Gödel
- Alfred Tarski
- Saharon Shelah

Philosophy portal

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